

SDVN Enabled Traffic Light Cooperative Framework for E-SIoV Mobility in a Smart City Scenario

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Abstract—The Social Internet of Vehicles (SIoVs) is an advanced approach to vehicular networking that connects specialized vehicles to share and exchange information, such as traffic jams, parking spaces, and more. This work aims to establish a network of emergency SIoVs (E-SIoVs) for frequent communication during vehicle movement across intersections to reduce average occupancy and waiting time. To achieve this objective, this work presents a novel software-defined vehicle networking-based E-SIoV framework assisted by a Smart Traffic Light Controller (STLC). Software-Defined Vehicular Networking (SDVN) based on an E-SIoV network utilizes communication between Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) for data receiving and control signal dissemination. The SDVN module generates a congestion prevention signal in the form of a new cycle phase duration. To do this, a modified linear-quadratic regulator, Quantize-LQR (QLQR), is proposed and deployed on SDVN-Controller (SDVNC). Following the output of QLQR as feedback, the proposed max-pressure-based STLC generates effective control signals. Further, two levels of prioritization are implemented in the city to improve E-SIoV movement: i) E-SIoV and ii) lanes with E-SIoV. In the first level, higher weight is assigned to E-SIoV, and in the second level, lanes are prioritized based on the quantity of E-SIoVs. A well-known open-source Simulation of Urban MOBility (SUMO) simulator has been utilized to validate the proposed framework on an Indian city's OpenStreetMap. The proposed framework outperforms other state-of-the-art approaches on several performance metrics.

Index Terms—Linear quadratic regulator (LQR) control, scheduling, social internet of vehicles, software-defined vehicular networking, traffic light controller.

I. INTRODUCTION

VEHICULAR technology has been enriched with several technologies, including smart sensors, the Internet of Things (IoT), computing/communication technologies, Road Side Units (RSUs), On-Board Units (OBUs), Intelligent Vehicles (IVs), and connected smart traffic lights in the last few decades [1], [2]. A hybrid network comprising Vehicular Ad-hoc Networks (VANETs), IoT devices, RSUs, and OBUs is emerging in the form of the Social Internet of Connected

Vehicles (SIoV) [3]. SIoV is a modern networking approach to IoV in which entities socialize with one another by sharing information of mutual interest [4]. Several intelligent services have been offered through SIoV, including on-road traffic, traffic information dissemination for nearby hospitals, and a wide variety of other utility services [5]. Moreover, SIoV networks address the safety issue of dynamic transportation by combining battery-powered intelligent devices, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and cloud/fog-based services [6]. SIoV is a vehicle-based cyber-physical system that can operate in tandem with IoT, VANETs, and Fog or Edge servers [7]. SIoV is a pioneering, exciting, and rapidly growing research domain in industrial and academic circles, with many opportunities for improving transportation infrastructure and vehicle networks. In this work, SIoV network is formed among Emergency SIoVs, i.e., E-SIoVs, and is managed by the Software Defined Vehicular Network (SDVN) architecture to facilitate the movement of emergency vehicles (ambulances, fire brigades, etc.) by minimizing traffic congestion in the city. Further, it has been observed that to reduce traffic congestion and improve traffic flow at the intersections for SIoV, Smart Traffic Light Controllers (STLCs) play a significant role. Therefore, this work proposes a novel STLC cooperative E-SIoV framework enabled by SDVN control.

A similar architecture was introduced by Sadio et al. [8], with four layers: fog, cloud, SDVN controller (SDVNC), and vehicles. This architecture offers programmable control, flexibility, scalability, and a global perspective. SDVN-enabled control systems have also been proposed for managing emergency traffic in several smart city applications [9]. Xia et al. [10] proposed an incentive scheme for SIoV networks based on fog computing of 2-dimensional behavior markers. The researchers evaluate vehicular nodes' behaviors and promote data-forwarding co-operation among selfish nodes. Further, in response to emergency circumstances, Li et al. [11] have recently proposed an emergency information diffusion model for SIoV. Based on this study, it is observed that existing technologies such as V2I, V2V, and VANETs empower the SIoV. However, several state-of-the-art research gaps for the E-SIoV framework need to be addressed as follows: i) Fog/Cloud has been used to offer compute and storage services in existing SIoV networks; however in heterogeneous vehicular networks, SIoV must be explored using SDVN-enabled architectures considering high priority emergency vehicles; ii) Due to the lack of congestion prevention signals generated by existing state-of-the-art methods, it is difficult to control traffic flow efficiently by detecting E-SIoV

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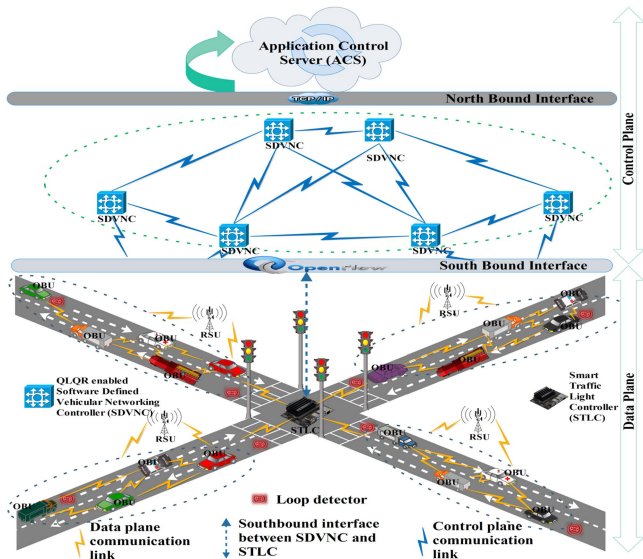


Fig. 1. Cyber-physical world interactions in SIOV environment illustrating hierarchical network architecture with SDVN environment.

on neighboring intersections' outgoing links in advance; iii) A traffic light cooperative SDVN architecture needs to be explored for solving the intersection congestion problem utilizing SIOV networks. This work proposes an SDVN-enabled and traffic light cooperative E-SIOV framework to fill the research gap mentioned earlier for city emergency services. The significant contributions of this work are enlisted as follows:

- This work proposes an SDN-enabled and traffic light cooperative architecture (Fig. 1) by decoupling the vehicular network into data and control planes for E-SIOV's movement in the city. In this, for smooth traffic flow on the road lanes with E-SIOV, two levels of prioritization are done: i) increasing the E-SIOVs weight factor value, and ii) increasing the lane priority by accounting the number of E-SIOVs.
- As a substitute for normal vehicle count, a heterogeneous vehicle weight assignment scheme is proposed to estimate an effective queue length while making an effective traffic light control signal decision making.
- To alleviate the congestion in the city, the SDVN-QLQR framework interacts with the RSUs and STLCs to monitor the traffic congestion periodically. Further, it generates the congestion prevention signal, for the STLC, in the form of new phase-cycle duration when congestion is detected at the intersection. Following this, STLC generates effective control signals to improve the traffic flow.
- An efficient algorithm is proposed by improving the Linear Quadratic Regulator (LQR) [12] i.e., Quantize-LQR (QLQR) to generate an optimal phase duration for STLC considering the congestion at the intersection, and it runs on SDVNC.
- Finally, a well-known open-source simulator (SUMO) [13] is utilized to conduct an extensive simulation study. In order to validate each module's significance, rigorous experimental analysis is conducted. Further, the proposed

SDVN-QLQR algorithm has been proven to be more efficient than other state-of-the-art algorithms on Average Waiting Time (AWT), Average Speed (AS), and Average Occupancy (AOCC) performance metrics.

The rest of this paper is organized as follows: The literature survey is presented in Section II in the field of Software-defined SIOVs to manage vehicular networks. Further, Section III presents system modeling, the SDVN-enabled framework, logical flow, and the solution approach. Section IV presents the simulation study and the experimental analysis of the proposed work, considering different state & input cost weighting matrices and comparison with same-line state-of-the-art methods. Finally, the conclusions and suggestions for future work are presented in Section V.

II. LITERATURE SURVEY

Several emergency services and safety applications have been suggested for the Social Internet of Vehicles (SIOVs), including traffic management, intelligent traffic surveillance, disaster management, and many more [4]. Consequently, an efficient architecture is indispensable for associating a single entity with multiple entities in SIOVs [14]. To meet the dynamic requirements of the vehicular network, software-defined SIOV has evolved into a vibrant, authentic, vigorous, and auto-configurable vehicular network to enable the vehicular network with programmable control. Bhatia et al. [15] proposed a more tractable and adaptable network setup to address the exponentially growing traffic in the wireless network via software-defined SIOV to improve efficiency. Oza et al. [16] introduced an optimal algorithm to maximize traffic flow while preventing spillbacks during normal operations and proposed an efficient strategy to recover from spillbacks caused by special events, such as blocked links due to disabled vehicles or emergency vehicle traversals. Rastgoftar et al. [17] presented a finite-state abstraction of traffic coordination and congestion in a network of interconnected roads (NOIR). The authors used Markov processes and Model Predictive Control (MPC) to optimize boundary inflow and traffic signal movement phases, effectively managing traffic congestion in NOIR simulations. Subsequently, Hosseinzadeh et al. [18] used MPC, offering both centralized and decentralized approaches to reduce congestion by prioritizing emergency vehicles. Several energy-efficient routing architectures were also proposed, considering the ability of software-defined SIOV to manage vehicular networks in multiple ways [19].

Alipio et al. [20] developed a software-defined test bed to accurately control traffic with value-based utility implementation and QoS provisioning in mind. Considering the need for a software-defined control network in ITS, several SDN-based control systems were designed for traffic management and control in urban emergencies [6]. Despite this, the authors failed to consider the congestion caused by traffic on the road lanes. Recently, Balta et al. [21] proposed SDN-based VANET (SDVN) and a 3-stage fuzzy-decision tree model for optimizing traffic signal optimization in large cities. The authors neglected the traffic heterogeneity and load balancing in congested city

areas. Liang et al. [22] proposed a deep reinforcement learning model by employing multiple optimization components, including target networks, double Q-learning networks, dueling networks, and prioritized experience replay. It is observed that the researchers did not consider vehicle priority during the agent's training and also did not consider real-time congestion control/monitoring. From the above study, it can be concluded that SDVN architecture has many advantages for the ITS in several aspects, such as real-time vehicle communication, congestion control/monitoring, etc. Further, SIOV is an emerging paradigm for providing services to the specific type of vehicles in the city. Furthermore, traffic light controllers also play a vital role in maintaining the city's smooth traffic flow. Thus, considering the above state-of-the-art and its associated characteristics, this work proposes a SDVN-enabled E-SIOV framework for emergency vehicles with minimal congestion flow with appropriate coordination of traffic light controllers.

III. THE PROPOSED SDVN E-SIOV FRAMEWORK

The proposed model comprises an SDVN-enabled traffic light cooperative E-SIOV framework that divides the vehicular network into the data and control planes, as depicted in Fig. 1. In this data plane, vehicles are equipped with OBUs. The OBUs send vehicle state information (ID, position, speed, etc.) to the RSUs through the V2I vehicular network using the IEEE 802.11p protocol. In the V2I network, RSUs are roadside infrastructures connected to the nearby STLCs and SDVNCs. The STLCs lie at the data plane and SDVNCs lie at the control plane. In the control plane, the SDVN controllers are connected to STLCs via the OpenFlow protocol using South Bound Interface (SBI), which is responsible for processing the received vehicular state data from the OBUs and generating the control signals for the traffic light controller signals in the form of new phase duration for the STLC (lies at the data plane). Following this phase duration, STLC generates the traffic light controllers following its main logic within the phase duration. Further, to update and maintain the SDVN controller and analyze the traffic data for future engineering, the Application Control Server (ACS) is connected to the SDVN Controllers via North Bound Interface (NBI) that can be formed using TCP/IP communication protocol.

A. System Modelling

In this work, the set of traffic light intersections and the set of road lanes in the city are modeled using graph theory. Therefore, the urban network is denoted with the help of an undirected graph $G=(V,E)$, where V represents the set of nodes (intersections comprised of road lanes), and E represents the set of road lanes. Each road lane is represented as an ordered pair $(p,q) \in E$. p and q denote a road lane's two end nodes (source and destination). The set of input nodes over the road lane (p,q) is represented by $I_{p,q}$, and the set of output nodes is represented by $O_{p,q}$. There are three types of nodes in this model: source nodes, intermediate nodes, and destination nodes. Source nodes are those nodes from where vehicles enter the network, intermediate nodes are the signalized intersection between source nodes, and destination nodes are the nodes from where traffic flows out of the network. The purpose of

this work is to propose a traffic light control system that generates green, yellow, and red traffic light signals at the intermediate nodes while reducing congestion at the intersections.

The model is discrete event-triggered, the average flow rates for exiting and entering at time step t on link (p,q) are represented by $\varrho_{p,q}^{exit}(t)$ and $\varrho_{p,q}^{enter}(t)$ respectively. The average traffic flow that enters and exits the road (p,q) , is modelled as

$$\varrho_{p,q}^{exit}(t) = \sum_{r \in O_{p,q}} \varrho_{p,q,r}^{exit}(t), \quad \varrho_{p,q}^{enter}(t) = \sum_{i \in I_{p,q}} \varrho_{i,p,q}^{exit}(t), \quad (1)$$

Here, t (a time slot) is the discrete-time index for the traffic network model and its controller. The average flow rates of the sub-stream directed towards the node r in link (p,q) at time step t , is represented as $\varrho_{p,q,r}^{exit}(t)$. The $\varrho_{p,q,r}^{exit}(t)$ is calculated as

$$\varrho_{p,q,r}^{exit}(t) = \min(\mu_{p,q,r}(t), \kappa_{p,q,r}(t), \vartheta_{p,q,r}(t)), \quad (2)$$

$$\mu_{p,q,r}(t) = \frac{\varpi_{p,q,r}(t) \times (\varsigma_{q,r} - n_{q,r}(t))}{T \times \sum_{i \in I_{q,r}} \varpi_{i,q,r}(t)}, \quad (3)$$

$$\kappa_{p,q,r}(t) = QL_{p,q,r}(t)/T + \varrho_{p,q,r}^{enter}(t), \quad (4)$$

$$\vartheta_{p,q,r}(t) = \varpi_{p,q,r}(t) \times \xi_{p,q} \times \nu_{p,q,r}(t)/T, \quad (5)$$

where, $\mu_{p,q,r}(t)$ represents the over-saturated flow rate of traffic conditions, $\kappa_{p,q,r}(t)$ represents unsaturated flow rate of traffic conditions, and $\vartheta_{p,q,r}(t)$ represents saturated flow rate of traffic conditions. Additionally, $\varpi_{p,q,r}(t)$ represents the lump of traffic which turns from (p,q) to (q,r) via an outward input, $\nu_{p,q,r}(t)$ represents the green signal time of the outgoing traffic flow from (p,q) towards r at time step t , $\xi_{p,q}$ represents the saturated flow rate which leaves (p,q) , and the capacity of the road (q,r) is denoted as $\varsigma_{q,r}$.

Referring to [23], the maximum capacity of road (p,q) is computed as

$$\varsigma_{p,q} = \frac{L_{p,q}}{L_{veh} + G_{\min}}. \quad (6)$$

where, $L_{p,q}$ is the road length, $G_{\min}$ is the minimal gap between vehicles, and L_{veh} is the average length of vehicles.

The number of vehicles in the queue, i.e., Queue Length ($QL_{p,q,r}$) is updated as

$$QL_{p,q,r}(t+1) = QL_{p,q,r}(t) + (\varrho_{p,q,r}^{enter}(t) - \varrho_{p,q,r}^{exit}(t)) \times T, \quad (7)$$

where, $\varrho_{p,q,r}^{enter}(t)$ is the flow rate towards r arriving at the end of queue, and it is given by

$$\varrho_{p,q,r}^{enter}(t) = \varpi_{p,q,r}(t) \times \varrho_{p,q}^{enter}(t), \quad (8)$$

$$\begin{aligned} \varrho_{p,q}^{enter}(t) &= (1 - \nu_{p,q}(t)/T) \times \varrho_{p,q}^{enter}(t - \varpi_{p,q}(t)) \\ &\quad + \nu_{p,q}(t)/T \times \varrho_{p,q}^{enter}(t - \varpi_{p,q}(t) - 1), \end{aligned} \quad (9)$$

$$TT_{p,q}(t) = (\varsigma_{p,q} - QL_{p,q}(t)) \times W_f^{veh}(t)/(k_{p,q}^{lane} \times SV_{p,q}^{free}), \quad (10)$$

$$\varpi_{p,q}(t) = \lfloor TT_{p,q}(t)/T \rfloor, \quad \nu_{p,q}(t) = TT_{p,q}(t) - \varpi_{p,q}(t) \times T. \quad (11)$$

where, $\varrho_{p,q}^{enter}(t)$ is the vehicles' flow rate arriving at the end of the queue on road (p,q) ; $TT_{p,q}(t)$ is the travel time required by a vehicle to arrive the end of the waiting queue on road (p,q) at time step t ; W_f^{veh} is heterogeneous vehicle's weight. The $k_{p,q}^{lane}$ is the road lanes' number on road (p,q) , and $SV_{p,q}^{free}$ is the free flow speed of the vehicles on road (p,q) [24]. Here, $\lfloor var \rfloor$ represents the smallest integer value not higher than var .

Following this, the maximum occupancy (OCC) of road (p,q) towards r is calculated as [13]

$$OCC_{p,q,r}(t) = \frac{TV_{p,q,r}(t)}{\varsigma_{p,q,r}}, \quad (12)$$

where, $TV_{p,q,r}(t)$ is the total number of vehicles at road (p,q) towards r , presented at time t . The Average Occupancy ($AOCC$) at any junction is estimated as

$$AOCC_{junction}(t+1) = \sum_{k \in nr} OCC_{p_k,q_k,r_k}(t). \quad (13)$$

where, nr is the total number of roads intersecting at any junction. The waiting time (WT) of the vehicles towards r is updated as

$$WT_{p,q,r}(t) = ((t - TT_{p,q}(t)/T) \times \varrho_{p,q}^{enter}(t) - \varrho_{p,q,r}^{exit}(t)) \times T, \quad (14)$$

Further, the Average Waiting Time (AWT) at any junction is estimated as

$$AWT_{junction}(t+1) = \sum_{k \in nr} \left(\frac{WT_{p_k,q_k,r_k}(t)}{TV_k} \right). \quad (15)$$

Based on this modeling, the obtained $AOCC$ and AWT are estimated to measure the quality of the solution.

B. SDVN Enabled Framework

This work considers a SDVN enabled E-SIoV framework consisting RSUs, intelligent vehicles with OBUs, SDN switches, and the SDVN controller. After deployment, all RSUs are stationary, and the vehicles are randomly positioned on the road lanes. At an intersection, vehicles can send information to their respective RSUs, while RSUs relay the information to associated SDVNCs after an appropriate data preprocessing. The SDVN controller can extract valuable traffic information received from RSUs and generates the phase duration and sends it back to the respective STLC. The set of vehicles represented by $Veh_{R_i}(t) = \{veh_1, veh_2, veh_3, \dots, veh_{|Veh_{R_i}(t)|}\}$, where $|Veh_{R_i}(t)|$ denotes total number of vehicles under the specific RSU at time t . The message consisting of several fields (Veh_{ID} , Veh_{speed} , Veh_{length} , Veh_{acc}) is broadcasted to a particular RSU at time t . The set of RSUs is denoted by $R_{SCon} = \{R_1, R_2, R_3, \dots, R_{|R_{SCon}|}\}$, where $|R_{SCon}|$ denotes total number of RSUs under a specific SDVNC ($SCon$). Each R_i ($1 \leq i \leq |R_{SCon}|$) has a set of messages, which is represented by $MRS_{R_i}(t) = \{mrs_{R_i}^1, mrs_{R_i}^2, mrs_{R_i}^3, \dots, mrs_{R_i}^{|MRS_{R_i}(t)|}\}$, where $|MRS_{R_i}(t)|$ is the overall number of unique messages sent by R_i at time t . The MRS_{R_i} contains message set as: $\{AWT, AOCC\}$. Each $mrs_{R_i}^j$ ($1 \leq j \leq |MRS_{R_i}(t)|$) refers to a data item in the RSUs message database which are required to

be disseminated in such a way that SDVNC associated with that junction/intersection receives the message. The set of SDVNCs denoted by $SDVNC_m = \{SDVNC_1, SDVNC_2, SDVNC_3, \dots, SDVNC_{|SDVNC_m|}\}$, where $|SDVNC_m|$ represents total number of SDVNCs. Each $SDVNC_m$ ($1 \leq m \leq |SDVNC_m|$) has a set of messages, which is represented by $MSS_{SDVNC_m}(t) = \{mss_{SDVNC_m}^1, mss_{SDVNC_m}^2, mss_{SDVNC_m}^3, \dots, mss_{SDVNC_m}^{|MSS_{SDVNC_m}(t)|}\}$, where $|MSS_{SDVNC_m}(t)|$ is the total number of unique messages sent by $SDVNC_m$ at time t . The MSS_{SDVNC_m} contains message set as: $\{AWT, AOCC, T\}$. The connections between RSUs to respective SDVNC, SDVNC to respective STLC, and SDVNCs to SDVNCs are known as link (l). Each link l has a bandwidth $BW_l(t)$, which is calculated as

$$BW_l(t) = To_l - BW_l^r(t). \quad (16)$$

where To_l is the overall bandwidth capacity of link l , and the flow rate of link l is denoted as $BW_l^r(t)$ at time t . There are F flow sets $FS^{(f)}$, $f = \{1, \dots, F\}$. Flow set between RSUs and SDVNC is identified by a tuple $\langle src_f, dst_f \rangle$, where src_f is the source node, and dst_f is the destination node. Flow set between SDVNCs and SDVNCs is identified by a tuple $\langle SDVNC_{src_f}, SDVNC_{dst_f} \rangle$, where $SDVNC_{src_f}$ is the source node, and $SDVNC_{dst_f}$ is the destination node. Each flow $f_s \in FS^{(f)}$ is identified by the 6-tuple: source/destination IP, source/destination port numbers, $|MRS_{R_i}(t)|/|MSS_{SDVNC_m}(t)|$, and transport protocol [25], i.e., the flow is recognised by both the tuple $\langle src_f, dst_f \rangle / \langle SDVNC_{src_f}, SDVNC_{dst_f} \rangle$ and its transport layer protocol. In a communication network, each flow $f_s \in FS^{(f)}$ should be routed through a single path $cp \in CP^{(f)}$, where $CP^{(f)}$ is the set of candidate paths for the flow set $FS^{(f)}$. The path set $CP^{(f)}$ contains paths that have $src_f / SDVNC_{src_f}$ and $dst_f / SDVNC_{dst_f}$.

The binary variable $\Lambda(t)$ is used for path selection result, which is selected as

$$\Lambda(t) = \begin{cases} 1, & \text{if flow } f \text{ selects transmission path } cp \in CP^{(f)}(t) \\ 0, & \text{otherwise} \end{cases}$$

At any given time, each flow will only be routed by one path, therefore the constraint exists as

$$\sum_{cp \in CP^{(f)}} \Lambda_{cp f_s}(t) = 1 \quad \forall f_s \in FS^{(f)}, f = \{1, \dots, F\}, \quad (17)$$

$$\Lambda_{cp f_s}(t) \in \{0, 1\} \quad \forall cp \in CP^{(f)} \quad \forall f_s \in FS^{(f)}, f = \{1, \dots, F\}, \quad (18)$$

Further, viewing the traffic demand $TD_{f_s}(t)$ of flow f_s at time step t , the path's bandwidth limitation cannot be surpassed:

$$\sum_{f_s \in FS^{(f)}} TD_{f_s}(t) \Lambda(t) \leq BW_{cp}(t) \quad \forall cp \in CP^{(f)}, f = \{1, \dots, F\}, \quad (19)$$

where, $BW_{cp}(t)$ is the unused bandwidth of path cp at time t . Traffic flows are routed by finding the best path between source and destination nodes. Firstly, we give the network utility function of flow set $FS^{(f)}$ prior to the formulation of the routing

problem:

$$U_f(t) = \sum_{f \in FS(f)} \sum_{cp \in CP(f)} \Lambda(t) [\alpha \times TH_{cp}(t) - \beta \times LT_{cp}(t)], \quad (20)$$

where, $TH_{cp}(t)$ is the throughput, and $LT_{cp}(t)$ is the end-to-end latency of path cp at time slot t , α and β are constant coefficients determined by specific requirements of traffic flows. As a result of this utility function, total throughput of all flows are maximized while end-to-end latency is minimized. After this, the routing problem can be formulated as:

$$\max \sum_{t=0}^T \sum_{f=1}^F U_f(t). \quad (21)$$

Considering the above utility function (20), all flows between RSUs to the nearby SDVNCs, SDVNCs to the STLCs, and SDVNCs to SDVNCs are formulated using (21) to maximize throughput subject to 18-20. In addition, the TCP/IP is used to send the messages between SDVNCs and ACS.

The formulated problem entails general constraints related to bandwidth (BW), the number of parallel connections (NC) [26], multi-hop communications ($MHCs$), and the packet length (PL), providing it a potentially non-convex problem [27]. Due to these diverse and unbounded constraints, multiple solutions with equivalent values can exist, making it challenging to attain the global optimal solution for non-convex problems [27]. To obtain the optimal solution, we have relaxed the constraints based on the proposed framework constraints, which optimize the utility function that confines the problem to a convex region. In the proposed framework, each RSU transmits vehicular information (AWT , $AOCC$) to its associated SDVNC using single hop communication, and each SDVN is directly connected to the nearby SDVNCs and its respective $STLC$ and transmits the information (AWT , $AOCC$) and congestion prevention signal (TQ) respectively by using one hop only. Further, considering the three or four-way intersection scenarios, we bound the number of parallel connections (NC) to a maximum of four. Additionally, in this framework, real-time communication occurs periodically between RSUs and the SDVN controller, as well as between the SDVN controller and $STLC$, with communication period synchronized to every phase duration cycle. From the above formulation, the objective function (21) focuses on throughput and latency, both of which are affected by variations in BW , NC , $MHCs$, and PL , where, Throughput ($TH_{t,cp}$) is calculated as $BW_{t,cp} \times NC_{t,cp}$, while Latency ($LT_{t,cp}$) is computed as $(\text{Number of hops} \times PL_{t,cp}) \div BW_{t,cp}$.

Since data transfer among RSUs, SDVNs, and STLCs occurs with single-hop communication. It implies that each flow set fs is relaxed to only one flow f and is transmitted via a single path cp only. Therefore, the constraint is relaxed as $\Lambda(t) = 1$ (17) and (18). Since $\Lambda(t) = 1$, (19) is also relaxed, considering $TD(t) \leq BW(t)$. Consequently, the utility function (20) is relaxed, considering $\alpha \times TH(t) - \beta \times LT(t)$ computed based on TH and LT , with equal weightage given to both the constant coefficients α and β . Following this convention, (21) gives a single value at the time step t .

C. Logical Flow of the Proposed Framework

The logical flow of the framework, SDVN-QLQR, is presented in Fig. 2. It begins by taking real-time vehicular dynamics as inputs. From inputs, *Real time Traffic Parameter Extractor()* module extracts the useful parameters such as vehicle type, Social Internet of Vehicle IDentity ($SIoVID$), Average Speed (AS) etc. With this, “*Lane Wise Heterogeneous Weight ($HWeight_{lane}$)*”, Average Waiting Time of lane (AWT_{lane}), Average Occupancy of lane ($AOCC_{lane}$) parameters are calculated. These parameters are, simultaneously, forwarded to the MP-TLC and SDVN components. SDVN component communicates between the “data plane” and the “control plane” by utilizing its message send/receive module. If Veh^{type} is $ESIoV$ then, $HWeight_{lane}$ is increased by γ , otherwise, $HWeight_{lane}$ is forwarded to “*Linearized System Parameter Matrices Calculator [A, B]*”. This module calculates the linearized system parameter matrices (A & B) and sends the calculated value to “*State feedback gain Matrix Generator (K)*”. This module generates the state feedback gain matrix (K) “using (22) and (23)”. Next, quantum value TQ is calculated by *Sum Of Diag(K)*, *Diag* function is used for extracting the diagonal values from state feedback gain matrix. If TQ is equal to 0 then it uses (24), else it uses (25) to update TQ . The generated TQ is forwarded to “*Phase Duration Calculator (PDC)*” through send/receive module of SDVN component. at the same time, PDC gets the output of “*Lane Pressure Calculator*”. Following this, the generated phase duration from the PDC is sent to the “*Set Phase Duration (SPD)*” module. SPD generates the phase duration and forwards it to “*Smart Traffic Light Controller ($STLC$)*”, which generates effective control signal (R/Y/G) as an output which improves the traffic flow at the intersection.

D. Algorithm

This sub-section presents the algorithmic description of the proposed framework.

Algorithm 1: It begins by taking certain inputs, as shown in Algorithm 1. In line 1, n is the loop variable ($0 \leq n \leq N - 1$). Next, in line 2, the time step for n^{th} junction, i.e., t_n is initialized to 0, and road lane variable k is also initialized to 0. In this work, T_n is fixed to 60 seconds which is the initial cycle phase duration for n^{th} junction. Next, if $(t_n \% T_n) == 0$, the algorithm enters in its execution loop for the maximum number of simulation steps. After that, in line 6, Social Internet of Vehicle Identity ($SIoVID_n(t_n)$) is generated using *getVehicleList(TV_n)*, where, TV_n is the total number of vehicles at n^{th} junction. Before going to further simulation steps, $HWeight$ of $SIoVID$ is calculated. The $HWeight$ of $SIoVID$ is to create a $W_{f_n}^{veh_s}$ for the respective junction to account for its vehicle heterogeneity. Based on the type of vehicles, $W_{f_n}^{veh_s}$ is updated as:

$$W_{f_n}^{veh_s} = \begin{cases} W_f^{pol}, & \text{if } Veh^{type} = Police \\ W_f^{mb}, & \text{if } Veh^{type} = MotorBike \\ W_f^{trk}, & \text{if } Veh^{type} = Truck \\ W_f^{emr}, & \text{if } Veh^{type} = Emergency \\ W_f^{car}, & \text{if } Veh^{type} = Car \\ W_f^{otyp}, & \text{Other types of vehicles} \end{cases}$$

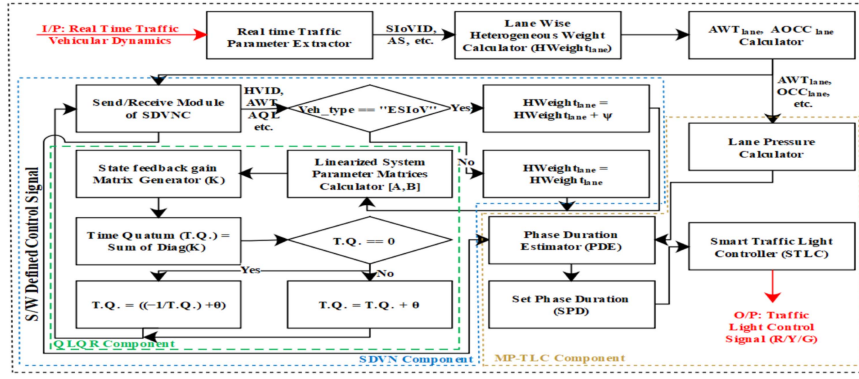


Fig. 2. Logical flow of the proposed SDVN-QLQR.

Algorithm 1: Main Algorithm.

Input: Total Number of junctions (N), Total Number of Vehicles (TV), Average Speed (AS), Vehicular Network of the city.

Output: Traffic light control signal (Green/Yellow/Red).

Notation:

t_n : Time Step for n^{th} junction

nr_n : Number of roads intersecting at n^{th} junction

s : {pol, trk, emr, car, mb, o_{typ} }

$W_{fn}^{veh_s}$: Weight of s type vehicles for n^{th} junction

lid : Lane index

```

1: for each  $n \in N$  do in parallel
2:   Initialize:  $k = 0, t_n = 0$ ;
3:    $T_n = 60 \triangleright$  Initial cycle phase duration for  $n^{th}$  junction
4:   while  $t_n < SimulationSteps$  do
5:     if  $(t_n \% T_n) == 0$  then
6:        $SIOVID_n(t_n) = getVehicleList(TV_n)$ ;
7:        $W_{fn}^{veh_s}(t_n) \leftarrow HWeight$  of  $SIOVID_n$ ;
8:       for each  $k \in nr_n$  do in parallel
9:         Compute  $AOCC_n(t_n)$  Using (13);
10:        Compute  $AWT_n(t_n)$  Using (15);
11:       end for
12:        $SndMsgToSDVNC(AWT_n, AOCC_n, T_n)$ ;
13:        $SDVNC(N, AWT_n, AOCC_n, T_n)$ ;
14:        $T_n = RecvMsgFromSDVNC(T_n(t_{new}))$ ;
15:        $STLC(N, AWT_n, AOCC_n, T_n(t_{pre}))$ ;
16:     end if
17:      $t_n = t_n + 1$ ;
18:   end while
19: end for

```

where, Veh^{type} is the vehicle type with its identity. The $W^{pol}, W^{mb}, W^{trk}, W^{emr}, W^{car}$, and $W^{o_{typ}}$ represent the weight factors of Police, MotorBike, Truck, Emergency (Ambulance and Fire_brigade), car and other types of vehicles respectively. The constants values are set based on experimental analysis, however, these values can be changed as per the application requirements. Following this, in lines 8 to 11, $AOCC$ and AWT are computed using (13) and (15), respectively. Next, traffic parameters are sent to SDVNC

using the utility function given in (21), which is represented as $SndMsgToSDVNC(AWT_n, AOCC_n, T_n)$. Based on these parameters, the SDVN-Controller calculates a new value of T_n that is received by the n^{th} junction. The definition for the same is given in Algorithm 2. Following this, $STLC()$ is called, the definition for the same is also given in Algorithm 3. Doing so, t_n is incremented by one. The same procedure is repeated for the number of simulation steps. The computational time of the proposed method (SDVN-QLQR) is approximately $2.348027979699691e+01$ seconds to complete the simulation. The average computational time for an episode is approximately $1.1183707e-02$ seconds.

Algorithm 2: In lines 1 to 3, SDVNC receives the traffic parameters of the nearby intersections (N_{nbr}) from each n^{th} STLC. In lines 4 to 8, if any lane is detected with the emergency vehicle (Veh^{emr}), the respective heterogeneous weight of that lane is automatically increased by adding the weight factor ψ . The value of ψ is decided based on various experimental test cases. With this, $AWT_{n,k}$ is inserted into the diagonal of matrix A as shown below:

$$A(t_n) = \begin{bmatrix} AWT_{n,0} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & AWT_{n,k} \end{bmatrix}$$

After that, $AOCC_{n,k}$ is then inserted into the diagonal of matrix B that is visualized as:

$$B(t_n) = \begin{bmatrix} AOCC_{n,0} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & AOCC_{n,k} \end{bmatrix}$$

Now, the system parameter matrices $A(t_n)$ and $B(t_n)$ are estimated as per lines 10-11 at each time step (t_n). Following this, the state feedback gain matrix $K(t_n)$ is obtained by solving the following non-linear differential Eq. of first order, i.e., Riccati [28], for matrices $S(t_n) = S^T(t_n) > 0$ and $K(t_n)$ simultaneously:

$$\begin{aligned} \mathbf{Q} + \mathbf{A}^T(t_n)\mathbf{S}(t_n)\mathbf{A}(t_n) - \mathbf{S}(t_n) - \mathbf{A}^T(t_n)\mathbf{S}(t_n)\mathbf{B}(t_n) \\ \mathbf{K}(t_n) = 0, \end{aligned} \quad (22)$$

Algorithm 2: SDVNC.

Input: N_{nbr} , AWT_n , $AOCC_n$, $T_n(t_{pre})$
Output: Updated cycle phase duration for each Junction T_n
Notation:
 $TQ(t_n)$: Quantize time
 $A \& B$: Linearized System parameter matrices(control input)
 lid : Lane index
 $Q \& R$: Diagonal Matrix $\triangleright$ State & Input Cost weighting matrix
 ψ : weight factor for emr vehicles

```

1: for  $n \in N_{nbr}$  do
2:    $ReceiveMsgFromTLC(AWT_n, AOCC_n, T_n(t_{pre}))$ ;
3: end for
4: for  $k \in nr_n$  do
5:   if  $Veh^{emr} \in vehicleType(SIoVID_{n,k})$  then
6:      $HWeight_{n,k}(t_n) = \psi + HWeight_{n,k}(t_n)$ ;
7:   end if
8: end for
9: for each  $k \in nr_n$  do in parallel
10:   $A(t_n).DiagAppend(AWT_{n,k})$ ;
11:   $B(t_n).DiagAppend(AOCC_{n,k})$ ;
12: end for
13: Calculate  $K(t_n)$  using (22) & (23);
14:  $TQ(t_n) = SumOfDiagK(t_n)$ ;
15: if  $TQ(t_n) == 0$  then
16:   Calculate  $TQ(t_n)$  using (24);
17: else
18:   Calculate  $TQ(t_n)$  using (25);
19: end if
20:  $T_n(t_{new}) = TQ(t_n)$ ;
21:  $SendMsgToSTLC(T_n(t_{new}))$ ;

```

$$K(t_n) = \mathbf{B}^T(t_n)\mathbf{S}(t_n)\mathbf{B}(t_n) + \mathbf{R}^{-1}\mathbf{B}^T(t_n)\mathbf{S}(t_n)\mathbf{A}(t_n). \quad (23)$$

Now, the quantize time $TQ(t_n)$ is calculated by adding all diagonal elements of a matrix $K(t_n)$. If $TQ(t_n) == 0$, then

$$TQ(t_n) = TQ(t_n) + \theta. \quad (24)$$

where, an extra value, i.e., θ is added as a correction factor. This factor can be kept different according to the architecture of the different junctions. Else,

$$TQ(t_n) = (-1/TQ(t_n)) + \theta. \quad (25)$$

Finally, the respective $T_j(t_{new})$ is sent to n^{th} TLC where it is to be used.

Algorithm 3: The objective of the function i.e., $STLC(N, AWT_n, AOCC_n, T_n(t_{pre}))$ is to generate the an effective phase duration for the respective road lane at the intersection. In lines 1–2, ϕ and χ values are set to 0.4 and 0.3 respectively [9]. However, these values can be changed as per the application requirements. In lines 3 to 5, $STLC$ triggers once it receives several traffic parameters i.e., AWT_n , $AOCC_n$, AS_n , and $T_n(t_{pre})$

Algorithm 3: STLC.

Input: N , AWT_n , $AOCC_n$, AS_n , $T_n(t_{pre})$
Output: Green/Yellow/Red Phase Duration.

```

1:  $\phi \leftarrow 0.4$ ;  $\triangleright$  Waiting time factor
2:  $\chi \leftarrow 0.3$ ;  $\triangleright$  Occupancy factor
3: for  $n \in N$  do
4:    $RecvMsgFromSDVNC(AWT_n, AOCC_n, AS_n, T_n(t_{pre}))$ ;
5: end for
6: for each  $n \in N$  do in parallel
7:   Calculate  $\mathbf{X}_{n,k}$  using (26);
8:    $[X_m, lid] = \max_{1 \leq n \leq k} (X_{n,k})$ ;
9:   for  $k \in nr_n$  do
10:     $SetGreenPhase(lid_k, \frac{X_{n,k} \times T_n}{\sum_{n \in N_{n,k}} X_{n,k}})$ ;
11:     $SetRedPhase(lid_d, \frac{X_{n,k} \times T_n}{\sum_{n \in N_{n,k}} X_{n,k}})$ ;  $\triangleright$  where,
         $1 \leq d \leq nr_n$ , and  $d \neq k$ 
12:     $SetYellowPhase(lid_{n,3}, 3)$ ;
13:   end for
14: end for

```

from SDVNC. Following this, in line 7, the pressure for k^{th} road lane of n^{th} junction ($X_{n,k}$) based on AWT and $AOCC$, is computed as follows:

$$X_{n,k} = \phi \times AWT_{n,k}(t_n) + \chi \times AOCC_{n,k}(t_n). \quad (26)$$

where, ϕ and χ are occupancy and waiting time factors respectively. Following this, the maximum pressure lane (X_m) initiates with the green phase duration, and the remaining lanes at the respective junction are set to the red phase for the same duration. Before switching between red to green or vice-versa, each lane traffic light uses 3 seconds for the yellow signal [29].

IV. SIMULATION STUDY

The performance behavior of the SDVN-QLQR controller is rigorously investigated by an extensive simulation study. The simulation is prepared on the Indian city OpenStreetMap utilizing SUMO simulator version 1.9.1 [13]. To analyze the effectiveness of SDVN-QLQR, several important performance metrics, i.e., Average Waiting Time (AWT), Average Speed (AS), and Average Occupancy (AOCC), are taken into account. AS represents the mean speed of vehicles at an intersection within a single phase duration. Further, the proposed ESioV framework is also compared with the several state-of-the-art traffic light controllers i.e., DQLC [9], MMC [30], LQRC [12], SDN-QMPC [29] to analyze its comparative performance.

DQLC [9] decides the duration of a traffic signal based on data generated by various vehicles and sensors with deep reinforcement learning. MMC [30] dynamically generates traffic light phase duration utilizing adaptive neuro-fuzzy based multi-mode intelligent TLC. LQRC [12] applied state-space equations to design an adaptive linear-quadratic regulator to minimize vehicles' average waiting time at the intersection. SDN-QMPC [29] adopts a quantize variant of max-pressure

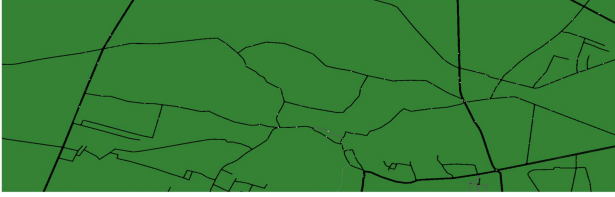


Fig. 3. Snapshot of the map used for the simulation study.

utilizing SDN. Further, the phase duration generated by SDN controller uses the fuzzy inference system (FIS) and sends it to respective TLC. From the results, it is proved that the quantize variant of max-pressure is a faster and more efficient method to work in a real-time scenario. In contrast, the proposed framework is built for E-SIoV with SDVN-enabled QLQR by proposing an improved MP-based STLC using occupancy and waiting time to estimate the road lane pressure. This STLC generates traffic light control signals, based on quantize time, received from the SDVNC by prioritizing E-SIoVs.

A. Simulation Parameters and Methodology

Python programming is used for implementing the proposed ESioV framework modules, and a well-known open-source simulator, i.e., Simulation of Urban MObility (SUMO) [13] has been used to simulate its vehicular network. For this study, an OpenStreetMap of Gwalior city (an Indian city) is downloaded as a .osm file using SUMO network conversion tool, i.e., netconvert; the same .osm file is changed into the vehicular network. The netconvert imports digital road networks from various sources and creates the vehicular network, i.e., road lanes, junctions/intersections, traffic lights, etc. Fig. 3 depicts the snapshot of the operating map created with SUMO.

B. Software Defined Vehicular Networking Component

There is a unique identification number assigned to every ESioV vehicle. These vehicles are detected using loop detectors, which are accounted in each lane. Particularly, emergency vehicles (Ambulance and Fire_brigade) are considered in this study. Although, ESioVs can be changed depending on application requirements in this framework. As one of the modules in SUMO, Traffic Control Interface (TraCI) provides the programming interface for controlling the SUMO traffic behavior utilizing Python. Table I shows the parameters in the SUMO simulation which have been used in this study. In SUMO simulation, randTrips.py script generates traffic on the vehicular road lane network based on OpenStreetMap. There are two approaches to generating traffic with this script: i) The departure (arrival/insertion) rate is controlled by option `-period <FLOAT>` (default one). Default settings generate vehicles with a constant period & departure rate of (one/period) per second, ii) By adding the option `-binomial <INT>`, the departure rate is randomized using a binomial distribution. The parameter INT represents the maximum number of simultaneous departures, which is given by the argument to `-binomial`. This study uses second approach to generate traffic by taking into

TABLE I
SUMO SIMULATION TRAFFIC PARAMETERS

SUMO Simulation Parameters	Value
Length of Car	4.5 m
Length of Police	3.5 m
Length of MotorBike	1.5 m
Length of Truck	8 m
Length of Emr-vehicles	7 m
Length of other vehicles	5 m
Max. Lane Speed	20 m/s
Max. Speed of Car	15 m/s
Max. Speed of Police	20 m/s
Max. Speed of MotorBike	13 m/s
Max. Speed of Truck	10 m/s
Max. Speed of Emr-vehicles	17 m/s
Max. Speed of other vehicles	13 m/s
Other Vehicle's Max Speed	13.9 m/s
Max. Acceleration of MotorBike	2.5 m/s ²
Max. Acceleration of Police	3.0 m/s ²
Max. Acceleration of Car	2.5 m/s ²
Max. Acceleration of Truck	0.8 m/s ²
Max. Acceleration of Emr Vehicles	1.5 m/s ²
Max. Acceleration of other vehicles	1.8 m/s ²
Max. De-acceleration of all Vehicles	4.0 m/s ²
Driving Proficiency(σ)	0.5
Detector Frequency	60
Min. Gap between Two Vehicles	2 m
Maximum Number of Lanes	4
Minimum Number of Lanes	2
Road lane type	Unidirectional
Binomial Distribution INT	2
Intersections Type (Y)	3 or 4-Way
Number of Traffic Light Junctions	16
Latitude of the map	26°9'42.0264'' N
Longitude of the map	78°4'21.3024'' E

account real traffic scenarios. To carry out this study, the value of INT is taken as 2, which can be changed as per the application requirements.

C. Results and Discussions

This sub-section portrays a detailed analysis of SDVN-QLQR's experimental findings and comparative study with other related works on various performance metrics.

1) *SDVN-QLQR Performance Behaviour on Varied State & Input Cost Weighting Matrix*: Several experiments are conducted on different parameter settings to assess the performance of SDVN-QLQR for other execution measurements. The behaviour of the SDVN-QLQR is highly dependent on the State & Input Cost weighting matrix, i.e., Q and R, which are the essential features of LQR. The matrices Q and R have to be symmetric positive semi-definite matrices [31]. When LQR is used, Q and R determine the optimal state feedback matrix K. K is mainly responsible for time domain response performance index [32]. Therefore, choosing the optimal weighting matrices Q and R are essential to design the optimal controller. This sub-section aims to test the proposed model on distinct parameters of Q and R to select an appropriate State & Input Cost weighting matrix. As a result, various performance metrics, i.e., Average Waiting Time (AWT), Average Speed (AS), and Average Occupancy (AOCC), are analyzed via a detailed investigation of various parameters of Q and R. Fig. 4(a)–(c) show the results

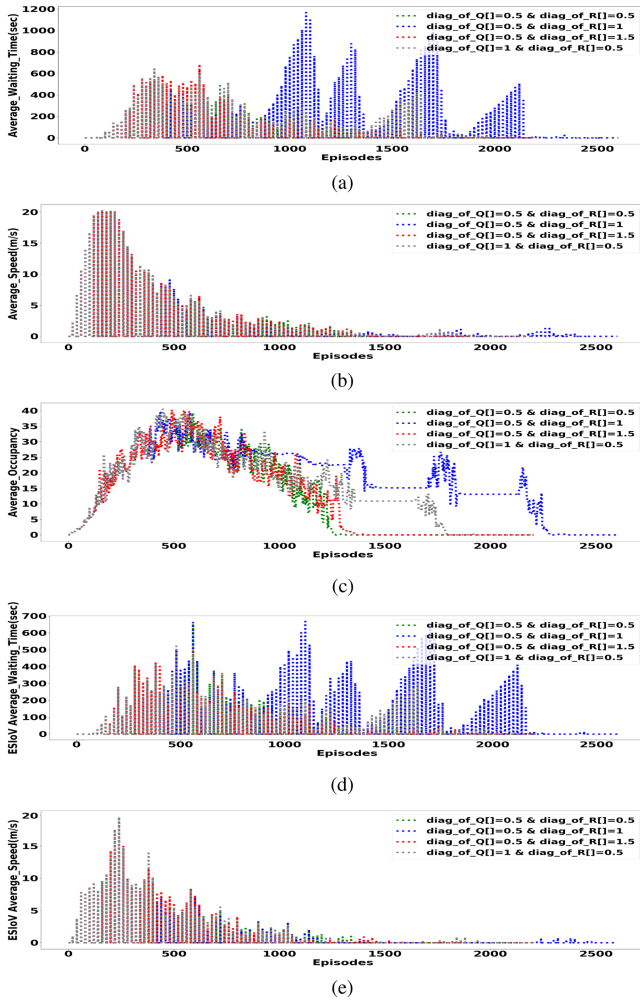


Fig. 4. State and Input Cost Weight parameter tuning comparison of SDVN-QLQR. (a) Results on average waiting time. (b) Results on average speed. (c) Results on occupancy. (d) Results on ESIoV average waiting time. (e) Results on ESIoV average speed.

corresponding to *AWT*, *AS*, and *AOCC* performance metrics, respectively, where episodes are shown on X-axis, and the performance metric is shown on Y-axis. Fig. 4(d) and (e) show the results w.r.t. E-SIoV on *AWT*, and *AS* performance metrics respectively. From Fig. 4(a), it is observed that *AWT* is minimal for $Q[] = 0.5$ and $R[] = 1.5$ scenario by approximately 15.8%, 58.9%, and 21.9% when compared to other adjustment scenarios, i.e., $Q[] = 0.5$ and $R[] = 0.5$, $Q[] = 0.5$ and $R[] = 1.0$, and $Q[] = 1$ and $R[] = 0.5$ respectively. The higher *AWT* value represents poor performance. Similarly, from Fig. 4(b), it is observed that *AS* is maximum for $Q[] = 0.5$ and $R[] = 1.5$ scenario by approximately 11.6%, 24.6%, and 19.7% when compared to other Q and R adjustments. Higher value of *AS* is preferred. Similarly, Fig. 4(c) shows the performance improvements of the *AOCC* when $Q[] = 0.5$ and $R[] = 1.5$ scenario is considered, then comparative performance improvement with other scenarios are 18.4%, 37.9%, and 21.9% observed. Lesser the value of *AOCC*, the less congestion at the intersection. Going further, Fig. 4(d) shows the performance improvements of the *AWT* w.r.t. E-SIoV when $Q[] = 0.5$ and $R[] = 1.5$ scenario is

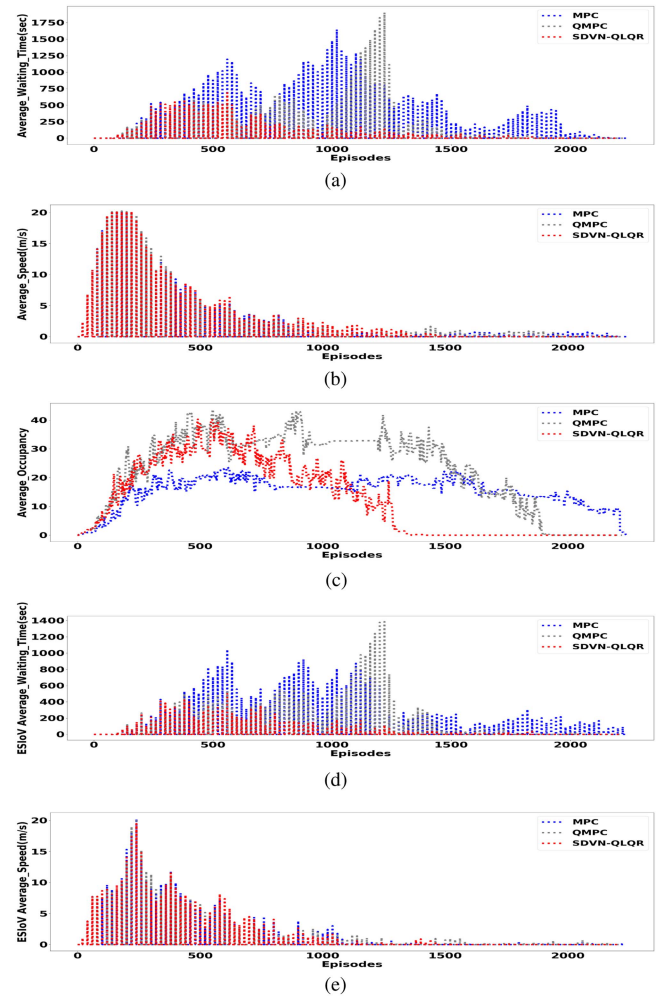


Fig. 5. Comparative results of SDVN-QLQR with MPC and QMPC. (a) Results on average waiting time. (b) Results on average speed. (c) Results on occupancy. (d) Results on ESIoV average waiting time. (e) Results on ESIoV average speed.

considered, the comparative performance improvement with other scenarios are 11%, 45.6%, and 21.1% observed. At last, Fig. 4(e) shows the performance improvements of the *AS* w.r.t. E-SIoV, when $Q[] = 0.5$ and $R[] = 1.5$ scenario is considered, then comparative performance improvement with other scenarios are 19.8%, 44.1%, and 23.9% observed. Higher value of *AS* indicates smooth traffic flow.

2) *SDVN-QLQR Comparative Analysis With QMPC and MPC*: SDVN-QLQR behaviour is also analyzed by conducting a set of experiments on different versions of MPC. In Fig. 5, legends represent the different versions of MPC, i.e., the proposed SDVN-QLQR, QMPC [29], and MPC [33]. Fig. 5(a)–(c) show the results corresponding to *AWT*, *AS*, *AOCC* performance metrics respectively and Fig. 5(d) and (e) show the results in terms of E-SIoV corresponding to *AWT*, and *AS* performance metrics. From Fig. 5(a), it is analyzed that SDVN-QLQR minimizes *AWT* w.r.t. QMPC and MPC by approximately 32.9% and 51.4%, respectively. Further, the Average Speed (*AS*) is determined in Fig. 5(b). From Fig. 5(b), it is observed that SDVN-QLQR maximizes *AS* by approximately 15.4% and

22.6% in comparison to QMPC and MPC, respectively. Similarly, Fig. 5(c) corresponds to the performance improvements of SDVN-QLQR for (*AOCC*), SDVN-QLQR improves 19.5% and 44.5% over QMPC and MPC, respectively. Further, for E-SIoV, the comparative analysis of SDVN-QLQR with QMPC and MPC on AWT and AS is shown in Fig. 5(d) and (e), respectively. From Fig. 5(d), it is observed that SDVN-QLQR beats QMPC and MPC in terms of *AWT* by 39% and 49.7%, respectively, considering E-SIoV scenario. From Fig. 5(e), it is also observed that SDVN-QLQR reaches the maximum speed value first in comparison to QMPC and MPC. It is analyzed from Fig. 5(e) that SDVN-QLQR performs 11% and 28.2% better than QMPC and MPC over *AS* metric. Fig. 5 illustrates that the traffic flow is smoother throughout all simulation episodes as an analysis of the results. As an outcome of incorporation of SDVN and quantize components, the proposed SDVN-QLQR model significantly improves the performance over different versions of MPC.

3) *Comparative Study*: To analyze the comparative performance behavior of SDVN-QLQR with the diverse nature of state-of-the-art algorithms, a set of experiments are done, and the outcomes are analyzed in this sub-section. To do this, MMC [30], LQRC [12], DQLC [9], and SDN-QMPC [29] are opted. The outcomes of the comparative results are presented in Fig. 6. For the *AWT* metric; it is noticed from Fig. 6(a) that performance improvement of SDVN-QLQR is 39.2%, 43.9%, 31.4%, and 23.5% over DQLC, MMC, LQRC and SDN-QMPC algorithms, respectively. From Fig. 6(b), it is understood that SDVN-QLQR is able to clear the traffic first and achieves the maximum *AS*. It depicts that the *AS* performance of SDVN-QLQR is better than DQLC, MMC, LQRC, and SDN-QMPC algorithms by 29.4%, 33.7%, 23.3%, and 19.7%, respectively. In the same line, it can be examined by Fig. 6(c) that the performance improvement of SDVN-QLQR for *AOCC* metric over DQLC, MMC, LQRC, and SDN-QMPC algorithms is 32.4%, 39.6%, 27.8%, and 19.1% respectively. After that, for E-SIoV, AWT and AS are improved significantly, as shown in Fig. 6(d) and (e), respectively. From Fig. 6(d), it is analyzed that SDVN-QLQR performs 31.8%, 36.5%, 23.3%, and 16.1% better than DQLC, MMC, LQRC, and SDN-QMPC algorithms on *AWT* metric in terms of E-SIoV. At last, Fig. 6(e) indicates that the SDVN-QLQR performs 20.7%, 24.8%, 14.1%, and 10.9% better than DQLC, MMC, LQRC, and SDN-QMPC algorithms for *AS* metric. The results show that with noteworthy improvements in *AWT*, *AS*, and *AOCC*, SDVN-QLQR has outperformed the state-of-the-art methods. The SDVNC monitors the congested routes and generates efficient phase duration based on expected congestion scenarios to attenuate intersection congestion. Using this phase duration, the traffic flow is regulated by the traffic light controller. As an inference, the SDVNC contributes in smooth traffic flow by generating an efficient phase duration, and it becomes a candidate add-on to the traffic light controller to optimize the congestion and performance metrics i.e., *AWT*, *AS*, and *AOCC* significantly.

4) *Comparative Study of Variance*: We have included three more metrics namely Waiting Time Variance (WTV), Speed Variance (SV), and Occupancy Variance (OV) [34], [35]. These

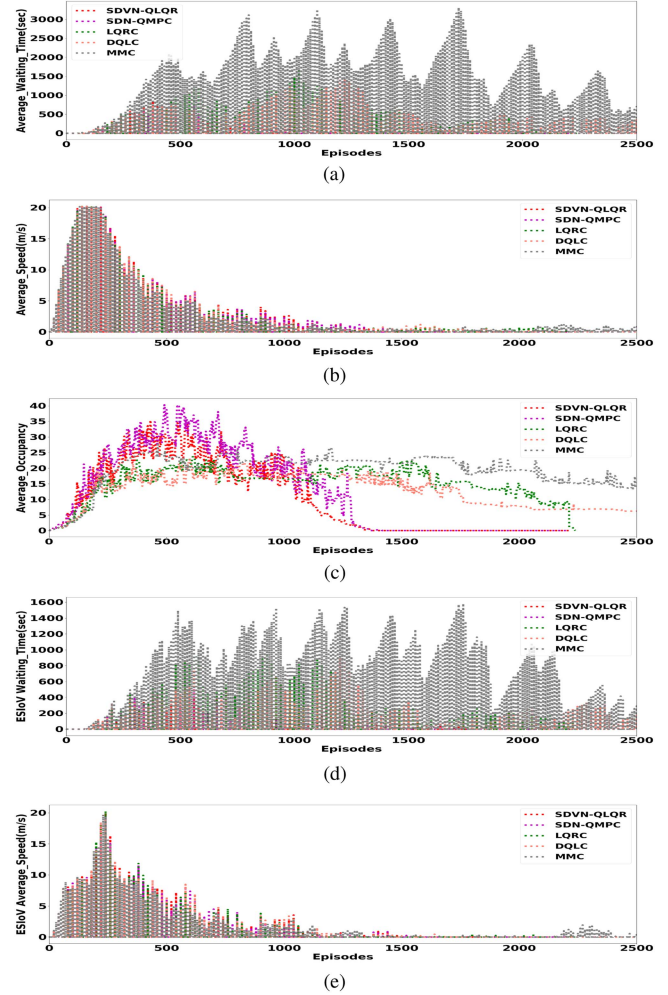


Fig. 6. Comparative results of SDVN-QLQR with other state-of-the-art algorithms. (a) Results on average waiting time. (b) Results on average speed. (c) Results on occupancy. (d) Results on ESioV average waiting time. (e) Results on ESioV average speed.

TABLE II
COMPARISON OF VARIANCE IN WAITING TIME, SPEED, AND OCCUPANCY
AMONG VARIOUS ALGORITHMS

Performance Matrix /Algorithm	SDVN-QLQR	SDN-QMPC	LQRC	DQLC	MMC
Waiting Time	2.56	2.85	7.87	11.02	31.7
Speed	2.24	2.94	8.75	17.86	37.76
Occupancy	2.86	3.97	6.41	12.48	28.46
E-SIoV Speed	1.2	2.28	5.79	12.77	19.61
E-SIoV Waiting Time	1.04	3.05	5.94	14.89	28.35

metrics are used to show the fairness of proposed algorithm. The results have been displayed in Table II. It is observed that the proposed algorithm consistently exhibits lower variance in these metrics, indicating reduced variability and bringing individual data points closer to their respective means. This reduction in variance signifies a notable reduction in the variation of waiting time (WT), speed (S), and occupancy (OCC) of vehicles across

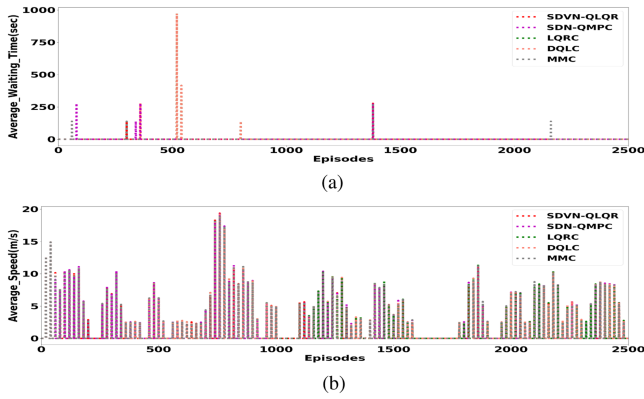


Fig. 7. Comparative results of SDVN-QLQR with other state-of-the-art algorithms utilizing the real-time traffic flow data [36]. (a) Results on average waiting time. (b) Results on average speed.

all the lanes at the intersection. The lower variance supports the validity and effectiveness of the proposed metrics, namely, average waiting time (AWT), average speed (AS), and average occupancy (AOCC).

5) *Comparative Study Using Real Traffic Flow Data:* A set of comparative experiments with several state-of-the-art algorithms has been done to analyze the comparative performance behavior of SDVN-QLQR considering the real-time traffic flow data [36]. For the comparative study, MMC [30], LQRC [12], DQLC [9], and SDN-QMPC [29] are used. Fig. 7 presents the results using the real-time data set captured by drone [36] in the simulation. For the AWT metric, it is observed from Fig. 7(a) that performance improvement of SDVN-QLQR is 44.48%, 31.60%, 39.43%, and 23.62% over MMC, LQRC, DQLC, and SDN-QMPC, respectively. Fig. 7(b) presents the results for the AS; from the results, it is inferred that SDVN-QLQR is able to clear the traffic first and achieves the maximum AS by improving its performance over MMC, LQRC, DQLC, and SDN-QMPC with 34.61%, 25.34%, 31.10%, and 20.83% respectively. The above results show that the proposed SDVN-QLQR has outperformed with other state-of-the-art algorithms in AWT and AS performance metrics considering the real-time traffic flow data [36].

V. CONCLUSION

This work introduces a SDVN enabled traffic light cooperative framework for E-SIoV mobility in a smart city scenario. In this framework, vehicles heterogeneity is accounted by assigning distinct weights. Further, E-SIoV and road-lanes are prioritized as E-SIoV are detected on the lanes. The SDVN module of the proposed framework periodically monitors congestion at the intersections and generates new phase cycle duration for the smart traffic light controller. Utilizing this phase duration as feedback, the proposed improved max-pressure traffic light controller generates control signals to improve the traffic flow. The framework's effectiveness is evaluated utilizing an open-source simulation tool (SUMO) on an Indian city. This work shows that the SDVN-QLQR is more effective than deep reinforcement learning, adaptive neuro-fuzzy, linear quadratic regulator-based

algorithm, and maximum pressure-based algorithms. In future work, a hardware test-bed prototype will be developed to analyze the communication latency between layers with signal loss. Following this, several social internet of vehicle services will be explored with the proposed framework.

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